

Comparison of Transillumination and Histologic Slide Measurements of Tumor Diameter in Uveal Melanoma

LELA POLIVOGIANIS, MS,¹ JOHANNA M. SEDDON, MD,^{1,2} ROBERT J. GLYNN, ScD,¹ EVANGELOS S. GRAGOUDAS, MD,² DANIEL M. ALBERT, MD³

Abstract: Tumor size is an important prognostic factor for deaths due to uveal melanoma. The objectives of this study are: (1) to determine the association between largest tumor diameter assessed by clinical transillumination and largest tumor diameter assessed by measurement of the histologic slide; (2) to predict a clinical transillumination diameter for each enucleation specimen based on the best combination of histologic variables—diameter, height, location, and tumor shape; and (3) to use the predicted transillumination diameter in place of histologic diameter measurement as a prognostic factor for patients treated by enucleation in order to reevaluate previously reported survival outcomes for patients treated by proton beam and enucleation. Comparison of measurements on 40 eyes indicated that transillumination largest diameter measurements were larger than histologic slide measurements (mean difference, 2.87 mm). Multiple linear regression analysis yielded the following equation: transillumination diameter = $[4.73 + (0.58 \times \text{histologic diameter}) + (0.38 \times \text{histologic height}) + (1.87 \times \text{anterior location})]$. Using the actual and predicted transillumination measurements, the results of this study support a previously published report on these data: treatment type, tumor size, and location were predictive of melanoma-related deaths, metastases, and all deaths. [Key words: enucleation, histology, prognostic factors, proton beam, irradiation, transillumination, tumor diameter, tumor height, tumor location, uveal melanoma.] *Ophthalmology* 95: 1576–1582, 1988

Previous studies have consistently shown that tumor size is an important prognostic factor for survival of pa-

tients with uveal melanoma after treatment.^{1–6} Patients with large tumors have a worse prognosis. Depending on the type of therapy selected, tumor size can be evaluated by a variety of techniques. Among the different size de-

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¹ Epidemiology Unit, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

² Retina Service, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

³ David G. Cogan Ophthalmic Pathology Laboratory, Massachusetts Eye and Ear Infirmary, Harvard Medical School, Boston.

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Reprint requests to Johanna M. Seddon, MD, Massachusetts Eye and Ear Infirmary, 243 Charles St, Boston, MA 02114.